

Rushil Shetty

<https://github.com/rushilss> | rushilrshetty@gmail.com | 267-614-6915

EDUCATION

University of Maryland

College Park, MD

Bachelor of Science, Computer Science

- Relevant Coursework: Object-Oriented Programming, Computer Systems, Machine Learning, Data Science, Bioinformatics, Data Structures, Algorithms, Data Visualization, Natural Language Processing
- Awarded UMD President's Scholarship for 4 years

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript, SQL

Frameworks & Tools: Git, Linux, Pandas, Numpy, PostgreSQL, Docker, GDB, FAISS, OpenAI, Tableau

Other Skills: RESTful APIs, ETL/ELT Data Pipeline, Artificial Intelligence, Agile, Cloud (AWS), RAG

RELEVANT EXPERIENCE

Computer Science Tutor - Doylestown, PA

Sept. 2024 - pres

- Delivered one-on-one tutoring sessions for local high school and college students enrolled in introductory computer science/Java courses, helping students achieve an average grade of 91% collectively

PROJECT EXPERIENCE

AI Movie Recommendation Tool – Python, RAG, Prompt Engineering, Artificial Intelligence: ([Github](#))

- Developed a movie recommendation system using Retrieval-Augmented Generation (RAG) and OpenAI's GPT-3.5-turbo to provide personalized movie suggestions based on user input.
- Integrated multi-query retrieval to enhance recommendation accuracy by generating additional context-aware queries, improving the system's ability to understand user intent and deliver suggestions.
- Processed a dataset of 250+ movies to generate recommendations and utilized Utelly API to provide information on streaming availability for recommended movies.

NCAA March Madness Optimal Bracket Generator - Python, Data Engineering: ([Github](#))

- Designed a predictive NCAA March Madness bracket generator utilizing game theory concepts and data scraped, cleaned, and merged from ESPN & FiveThirtyEight.
- Achieved a 99th percentile bracket accuracy in 2022 and 85th percentile in 2023 through statistical analysis and data-driven insights to optimize bracket predictions.
- Implemented algorithms to optimize selection based on field leverage and team statistics

Interactive World Data Visualization – Javascript, D3.js, Data Visualization: ([Youtube Demo](#))

- Developed an interactive GDP visualization tool using D3.js, enabling users to explore GDP data for 195+ countries with real-time filtering and dynamic tooltips for improved data representation.
- Achieved a 40% reduction in user interaction time by implementing smooth transitions and enhancing accessibility features for non-technical users, improving overall user experience.

FM Index for Genomic Data Querying – Java, Data Structures, Algorithm Optimization:

- Implemented an FM Index and suffix array algorithms to enable fast genomic sequence searching, enabling efficient analysis of genomic datasets spanning 1000+ base pairs of DNA
- Optimized the FM Index construction process, reducing memory usage by 25% and improving query response times by 15% for datasets exceeding 1000 base pairs for scalability

MIPS 5-Stage Pipelined Processor Simulation – C, Systems Architecture:

- Designed and implemented a five-stage MIPS processor pipeline (IF, ID, EX, MEM, WB), achieving a reduction in instruction execution time using data forwarding and hazard detection